

STANDARD OPERATING PROCEDURE

SOP-001

Hard Drive & Laptop Forensic Imaging and Acquisition

Digital Forensics Training Programme — Module 1: Core Acquisition Competency

Document Control	
Document ID	SOP-001-v1.2
Status	Draft — for review / training use
Owner	Digital Forensics Team Lead
Applies to	All trainee and junior forensic examiners
Prerequisite	SOP-000: Chain of Custody & Exhibit Handling (must be signed off first)
Gates progression to	SOP-002: Forensic Image Verification & Processing
Review cycle	Every 12 months, or on tool/version change
Classification	Internal — Confidential

1. Purpose

This SOP defines the mandatory, minimum-standard procedure for the ACQUISITION PHASE of digital evidence from a laptop or desktop computer's internal storage — from the point the exhibit is presented for imaging, to a verified forensic image with its acquisition log. It is a core competency gate in the training programme: a trainee must demonstrate this procedure correctly, under supervision, before progressing to later SOPs covering artefact processing and reporting.

The objective of every acquisition is to produce a complete, verifiable, forensically sound copy of the source media while altering the original evidence as little as technically possible, and to fully document every decision and action taken.

CHAIN OF CUSTODY — COVERED SEPARATELY (SOP-000)

- Chain of custody and exhibit handling procedures (continuity forms, exhibit labelling, secure storage, movement logging) are defined in SOP-000: Chain of Custody & Exhibit Handling.
- SOP-000 MUST be completed and signed off before a trainee begins this SOP.
- Where this document says “record on the exhibit/continuity record” or “per SOP-000”, follow the procedure in that document — it is not repeated here.

2. Scope

This SOP covers the acquisition phase only:

- Initial triage of a computer/laptop prior to acquisition.
- Decision-making for live (running) vs dead-box (powered off) systems.
- Handling of encrypted volumes, RAID/multi-disk configurations, and volatile (RAM) data.
- Physical drive removal and imaging via hardware write blocker.
- In-situ / software acquisition where drive removal is not possible.
- Use of standalone hardware forensic duplicators.
- Image verification: hashing, acquisition log review, and write-blocker discipline.

Out of scope: chain of custody / exhibit handling (SOP-000), mobile device acquisition, cloud/remote acquisition, memory forensics analysis, and file system/artefact examination — these are covered in separate SOPs.

COMPETENCY GATE

- SOP-000 (Chain of Custody & Exhibit Handling) must be signed off before starting this SOP.
- Trainees must not perform an unsupervised acquisition on live evidence until signed off against Section 11.
- All practice runs must use training/test media only — never live case evidence — until sign-off is complete.

3. Authority, Legal Basis & Health and Safety

- Confirm written authorisation/warrant/consent covers the device before touching it. If in doubt, stop and escalate to the Team Lead.
- Photograph the device, all cabling, ports, and the surrounding scene in situ before any interaction, and record the exhibit per SOP-000.
- Use anti-static precautions (wrist strap / mat) when opening a chassis or handling bare drives.

- Never connect a suspect drive directly to a workstation without a write blocker in place.
- If a device appears to be an active, powered cryptocurrency wallet, POS terminal, or server with live client impact, escalate before proceeding — special handling applies.

4. Decision Tree Overview

Every acquisition begins by working through the two decision trees below in order. Figure 1A resolves the power state, volatile data, and encryption triage. Figure 1B resolves the physical acquisition method once the device is safely at rest (or confirmed dead-on-arrival).

A small, faint diagram representing the initial triage decision tree for power state, volatile data capture, and encryption. The diagram is too small to be legible but is positioned centrally between the text and the caption.

Figure 1A — Initial triage: power state, volatile data capture, and encryption.



Figure 1B — Acquisition method decision tree for a dead-box / powered-down system.

5. Step-by-Step Procedure

5.1 Initial Triage & Documentation

1. Photograph the device as found: all sides, screen state, attached peripherals, port usage, and any visible damage.
2. Record make, model, serial number, asset tag, and exhibit/case reference on the exhibit/continuity record per SOP-000.
3. Determine power state: is the screen on, are fans/drive activity lights running, is there any display output?
4. Proceed to Figure 1A.

5.2 If the Device is Powered On (Live System)

5. Do not close the lid, log off, or restart under any circumstance before completing this step.
6. If the OS is unlocked or the password is known, assess whether volatile data capture is required (usually yes for active investigations — running processes, network connections, logged-in users, open encrypted containers).
7. Capture RAM using a trusted, verified tool run from external write-protected media (FTK Imager or Magnet RAM Capture). Never install software onto the suspect drive.
8. Check encryption status: look for BitLocker, FileVault, VeraCrypt, LUKS, or a self-encrypting drive (SED). Where possible, export or screenshot the recovery key (e.g. via manage-bde, or the organisation's Active Directory/Azure AD escrow) while the system is unlocked.
9. If encryption is present and no recovery key can be obtained, you must perform a live logical or full-disk image before the system is powered down — shutting down first will make the data unrecoverable without the key.
10. If no encryption is present, or the key/recovery password has been secured, perform a controlled, forensically-aware shutdown and proceed to Section 5.3.

5.3 If the Device is Powered Off (Dead Box)

11. Confirm the device is safe to briefly power on for a boot/health check (no signs of physical damage, liquid ingress, or burning smell).
12. If it will not power on safely, do not force it. Document the fault and escalate per Section 5.6.
13. If it powers on safely, check whether the drive is part of a RAID or multi-disk array (inspect BIOS/UEFI, controller card, or physical bay layout).
14. For RAID: document the controller make/model, RAID level, disk order, and stripe size before touching any disk. Where possible image each physical member disk individually, then reconstruct the array logically during processing (X-Ways, Magnet AXIOM). If the array cannot be split safely, image through the controller in place instead.
15. Determine whether the drive can be safely removed (standard SATA/NVMe drive bay) or whether it is soldered/BGA-mounted, behind a proprietary enclosure, or otherwise impractical to remove.

5.3.1 Drive Can Be Removed

16. Power down fully and remove the drive using appropriate tools and anti-static precautions.
17. Connect the drive to a forensic write blocker (hardware, not software-only) — e.g. Tableau or WiebeTech.
18. Connect the write blocker to the forensic acquisition workstation.
19. Image using FTK Imager or X-Ways Forensics, selecting E01 (EnCase) or raw/DD format per case requirements. Capture a full E01 case-info header (examiner name, case number, evidence number, notes).

5.3.2 Drive Cannot Be Removed / In-Situ Acquisition

20. Boot the target machine from a trusted forensic boot medium (e.g. Paladin, CAINE, or SIFT USB) that mounts internal drives read-only by default.
21. Alternatively, if the drive interface allows it, connect via a PCIe/NVMe/M.2-to-USB adapter routed through a hardware write blocker to the forensic workstation.
22. Image using FTK Imager or X-Ways Forensics as above.

5.3.3 Hardware Forensic Duplicator (Either Path)

23. A standalone duplicator such as a Tableau TX1 or TX2 may be used instead of a workstation + write blocker + software combination for either 5.3.1 or 5.3.2.
24. These devices perform write-blocked imaging, hashing, and verification independently, and are preferred in the field or where a forensic workstation is not available.

5.4 Image Verification

Verification confirms that the forensic image is a complete, unaltered, bit-for-bit copy of the source before the source drive is released and before any analysis begins. This is a mandatory step, not an optional extra — an unverified image is not considered forensically sound.

5.4.1 Write Blocker Requirements During Verification

25. Keep the source drive connected via the hardware write blocker for the entire acquisition and verification process — do not disconnect it early to "save time."
26. If verification requires re-reading the source drive (e.g. a second hashing pass, or resolving a mismatch), it must be re-read only through a hardware write blocker, never a direct connection.
27. If using a standalone hardware duplicator (e.g. Tableau TX1/TX2), confirm the unit's write-block status indicator/log shows the source was write-protected for the full duration of the job before accepting the result.
28. Never verify an image by mounting the source drive read-write "just to compare a few files" — this invalidates the acquisition.

5.4.2 Hash Verification

29. Generate both MD5 and SHA-256 hashes of the source drive and of the completed image.
30. Confirm the acquisition tool reports a hash match between source and image before the source drive is disconnected from the write blocker.
31. If hashes do not match, do not discard the failed image or the source drive. Stop, document the discrepancy, and re-image using a fresh target and, where possible, a different tool or write blocker to isolate the cause before escalating to the Team Lead.
32. Record both hash values on the exhibit/continuity record (per SOP-000) and in the case management system.

5.4.3 Acquisition Log Review

33. Every imaging tool (FTK Imager, X-Ways, or a hardware duplicator) produces an acquisition/audit log — do not close the tool until this log has been exported and reviewed.
34. Check the log for: start/end timestamps, source and destination device identifiers, sector counts read vs. expected, any reported bad sectors/read errors, and the final verification (hash match) result.
35. Investigate and document any bad sectors or read errors reported in the log, even if the overall hash still verifies — these can indicate a failing drive that should be prioritised for re-imaging or specialist handling.
36. Save the acquisition log alongside the image file (same case folder) and attach a copy to the case file per SOP-000 — the log is evidence of the verification, not just the hash values quoted from it.
37. Confirm the log's reported source drive serial number and capacity match what was physically recorded in Section 5.1 — this catches cabling/port mix-ups on multi-drive workstations.

5.4.4 Working Copy

38. Never work directly from the original (master) image file. Duplicate a working copy for analysis and re-verify its hash against the master before use.
39. Store the master image read-only where the file system/storage platform supports it.

5.5 Storage & Handover (see SOP-000)

40. Store the original media in an anti-static, tamper-evident bag, labelled with exhibit number, and return it to secure evidence storage per SOP-000.
41. Store the forensic image (working copy) on the case storage server.
42. Complete all continuity/exhibit records required by SOP-000 before leaving the acquisition workstation.

5.6 Escalation (Device Will Not Power On / Physical Damage)

43. Do not attempt chip-off, JTAG, or ISP (In-System Programming) extraction unless you are specifically trained and signed off for that competency.
44. Photograph and document the condition of the device and drive.
45. Escalate to the Team Lead or a specialist data-recovery/forensic hardware partner.

6. Equipment & Tool Reference

Recommended core and supplementary tooling for this SOP. Trainees should be familiar with at least one option in each category before sign-off.

6.1 Approved Imaging Software (Team Standard)

The team standard imaging tools are FTK Imager, X-Ways Forensics, and Magnet ACQUIRE. Trainees must be competent in FTK Imager (primary) and at least one of the other two before sign-off.

Tool	Role in this SOP	Notes
FTK Imager (Exterro)	PRIMARY: disk imaging, live logical acquisition, RAM capture, image preview	Runs portable from USB for live systems; E01/raw(dd)/AD1 output
X-Ways Forensics	SECONDARY: disk imaging, live acquisition, RAID reconstruction; full analysis suite	Fully portable, very fast, low footprint; E01/raw output; also used in later analysis SOPs
Magnet ACQUIRE	SECONDARY: disk/removable-media imaging, targeted acquisition	Simple guided workflow; good training tool; E01/raw output with logging

6.2 Software Feature Comparison

Capability	FTK Imager	X-Ways Forensics	Magnet ACQUIRE
Full physical (bit-for-bit) disk image	Yes	Yes	Yes
Logical / targeted acquisition	Yes (AD1)	Yes (containers)	Yes (targeted)
Live RAM capture	Yes	No (use FTK Imager)	No (use FTK Imager)
E01 output	Yes	Yes	Yes
Raw (dd) output	Yes	Yes	Yes
Hash verification (MD5/SHA)	Yes	Yes	Yes
Acquisition/audit log produced	Yes	Yes	Yes
Runs portable (no install)	Yes (Imager Lite)	Yes (USB stick)	Portable executable
Evidence preview before imaging	Yes	Yes	Limited
RAID reconstruction	No	Yes	No
Analysis capability beyond acquisition	Preview only	Full suite	No (acquisition only)

6.3 Software Licensing Requirements

Licensing status should be re-confirmed with the vendor before purchase or deployment, as terms change. Status below is as at the date of this document.

Software	Cost	Licence mechanism	Dongle required?
FTK Imager (standard)	Free	Free download from Exterro (registration form); no licence key	No
FTK Imager Pro	Paid — annual subscription per user	Electronic licence code from Exterro	No
X-Ways Forensics	Paid — perpetual licence + update maintenance	USB dongle (local or network) or BYOD (a user-supplied USB device acts as the key)	Yes — dongle or BYOD device must be attached to run
Magnet ACQUIRE	Free	Free download from Magnet Forensics (community registration)	No
Magnet RAM Capture	Free	Free download from Magnet Forensics (free tools page)	No
LICENSING PRACTICE NOTES • X-Ways dongles are NOT replaced if lost or stolen — treat the dongle as a controlled asset: log it in/out like an exhibit and store it securely. • Free tools (FTK Imager, Magnet ACQUIRE, Magnet RAM Capture) must still be downloaded only from the official vendor site, and the installer's hash verified and recorded before deployment to forensic workstations. • Record the exact tool name and version used in every acquisition log entry — version matters if the acquisition is later challenged.			

6.4 Hardware Write Blockers

Device	Interface	Notes
Tableau T35689 / T8-R2	SATA/IDE/USB	Standard bench write blocker
WiebeTech Forensic UltraDock	SATA/IDE/USB	Popular, durable field option
CRU WiebeTech Ditto	Standalone duplicator	Dock + duplication in one unit

6.5 Hardware Forensic Duplicators

Device	Notes
Tableau TX1	Compact, field-friendly standalone imager with hashing/verification and network acquisition support
Tableau TX2	Higher-throughput successor to TX1; faster imaging speeds, additional interfaces
Logicube Falcon-NEO	Alternative standalone duplicator, widely deployed in LE labs

6.6 Volatile Data (RAM) Capture Tools

Tool	Use
FTK Imager	Live memory (RAM) capture via Capture Memory function, run portable from external media
Magnet RAM Capture	Free, lightweight live memory acquisition with minimal footprint on the target system

6.7 Bootable Forensic Environments

- Paladin Edition (Linux, forensic-focused, read-only default mounting)
- CAINE (Computer Aided INvestigative Environment)
- SIFT Workstation (SANS)

6.8 Adapters & Consumables

- SATA / mSATA / M.2 (NVMe & SATA) / U.2 to USB3 adapters, ideally routed through a write blocker
- Assorted screwdriver/bit set (Torx, Pentalobe, tri-wing) for laptop chassis entry
- Anti-static bags, evidence tape, exhibit labels, wrist strap
- Spare forensically wiped target drives (verified zero-fill, logged) for image storage in the field

6.9 Suggested Additions for the Training Lab

- A small library of “broken”/edge-case training laptops: one BitLocker-locked, one RAID-1 laptop, one soldered-storage ultrabook, one with a dead motherboard but healthy drive.
- A UFED/Cellebrite touch or similar for later modules, to give trainees a consistent vendor ecosystem across computer and mobile evidence.
- A shared, version-controlled hash-verification log (spreadsheet or LIMS) so trainees get used to formal exhibit tracking from day one.

7. Annotated Imaging Walkthrough (Illustrative)

The figures below illustrate the three screens that matter most during a software acquisition, using a generic dialog layout common to FTK Imager, X-Ways, and Magnet ACQUIRE. They are illustrative diagrams, not vendor screenshots — when this SOP is finalised, replace each figure with a real screenshot captured from the team's licensed installation (Alt+PrtScn on the acquisition workstation, using training media).

7.1 Selecting the Source

Always select the physical drive (not a logical volume) unless the case specifically calls for a logical/targeted acquisition, and always confirm the drive shown is the one presented through the write blocker — check model, serial, and capacity against the exhibit record before clicking Next.

Figure 2 — Source selection (illustrative). The serial/capacity cross-check against the exhibit record happens here.

7.2 Image Type & Evidence Information

The team default is E01 with case information completed in full. The verify-after-creation option must never be unticked — it is what produces the hash comparison relied on in Section 5.4.

Figure 3 — Image format and evidence information (illustrative). Case info entered here is embedded in the E01 header.

7.3 Reading the Verification Result

A pass requires BOTH hash pairs (MD5 and SHA-256) to match AND the bad-sector count to be reviewed. Export the log before closing the dialog.

Figure 4 — Verification results screen (illustrative). This is what a pass looks like; any mismatch means stop and follow 5.4.2.

8. Quality Assurance Checklist

To be completed for every acquisition, live or dead box, and retained with the case file.

#	Check	Y/N
1	Authorisation confirmed before device was touched	
2	Scene and device photographed before interaction	
3	Power state correctly identified and documented	
4	Volatile data captured where required (or justified why not)	
5	Encryption status checked and recovery key secured where possible	
6	RAID/multi-disk configuration documented before disassembly	
7	Hardware write blocker used for all connections to source media	
8	Source and image hashes match (MD5 and SHA-256)	
9	Acquisition log exported, reviewed, and saved with the image (bad sectors/errors checked)	
10	Source drive kept behind a write blocker for the full acquisition and verification	
11	Continuity/exhibit records completed per SOP-000	
12	Original media returned to secure evidence storage	

9. Common Pitfalls for New Examiners

- Shutting down an encrypted, unlocked laptop before securing the recovery key — this can make the evidence permanently inaccessible.
- Connecting a suspect drive directly to a workstation's onboard SATA port without a write blocker, even “just to check.”
- Skipping RAID documentation and later being unable to reconstruct disk order.
- Forgetting to hash the source drive itself, and only hashing the resulting image file.
- Assuming a laptop with no visible drive bay has no removable storage — check for M.2 slots under the keyboard/base plate before concluding a chip is soldered.
- Using a personal or non-forensic USB drive as the imaging destination without first verifying it is wiped and logging its serial number.

10. Glossary

Term	Definition
Dead box	A device that is powered off at the time of seizure/acquisition.
Write blocker	Hardware or software device that permits read access to storage media while physically preventing any write commands from reaching it.
E01 / EWF	EnCase Evidence File format — a compressed, hashed, segment-able forensic image container.
Chain of custody	The documented, unbroken record of who has handled evidence, when, and why, from seizure to disposal. Procedures are defined in SOP-000.
SED	Self-Encrypting Drive — a drive with encryption built into its firmware/controller.
Chip-off	Physical desoldering of a memory chip for direct forensic extraction; last-resort, specialist technique.
JTAG / ISP	Low-level hardware interfaces used to extract raw data directly from a device's processor/flash memory.

11. Trainee Competency Sign-Off

To be completed by the supervising examiner once the trainee has demonstrated this SOP end-to-end on training media.

Field	Detail
SOP-000 (Chain of Custody) previously signed off (Y/N)	
Trainee name	
Trainer / assessor name	
Date of assessment	
Training media / scenario used	
Live acquisition demonstrated (Y/N)	
Dead-box drive-removal acquisition demonstrated (Y/N)	
In-situ / bootable acquisition demonstrated (Y/N)	
Hash verification demonstrated (Y/N)	
Acquisition log review demonstrated (Y/N)	
Continuity records completed correctly per SOP-000 (Y/N)	
Result	Pass / Not yet competent (recheck required)
Cleared to progress to SOP-002	Yes / No
Trainer signature	